

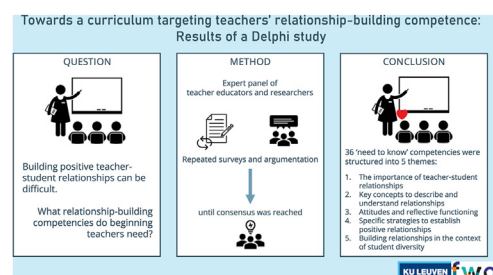


Research paper

Towards a curriculum targeting teachers' relationship-building competence: Results of a Delphi study

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GRAPHICAL ABSTRACT



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ABSTRACT

Although teachers often refer to personal relationships with students as a motivating and fulfilling part of their job, they also report that building these relationships is one of the most difficult aspects of teaching. This Delphi study investigated which competencies beginning teachers need to build relationships with individual students. Thirty-six competencies related to the importance of teacher-student relationships, key concepts, attitudes, specific strategies, and student diversity, were identified as 'need to know' by a panel of teacher educators and researchers. For most competencies, an intermediate level of mastery was expected. Results of the study can inspire teacher training.

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Teacher-student relationships are at the heart of education. Positive, warm relationships between teachers and students are beneficial both for the well-being and development of the student (e.g. Ansari, Hofkens et al., 2020; McGrath & Van Bergen, 2015; Nguyen et al., 2020; Roorda et al., 2014; Roorda & Koomen, 2021), and for the well-being of the teacher (e.g. Ansari, Pianta et al., 2020; Corbin et al., 2019; Hagenauer et al., 2015; Zee et al., 2017).

Teachers' ability to build a positive relationship with each of their students can thus be critical for both their own and their students' well-being. However, reports from pre-service and in-service teachers revealed that they consider interacting and building relationships with their students one of the most difficult aspects of teaching and a substantial number of teachers do not feel self-efficient in coping with conflictual teacher-student relationships (Borremans & Spilt, 2022; Jensen et al., 2015). Teacher-student relationships have been studied from multiple perspectives resulting in a rich academical knowledge base. However, it is less clear which of these scientific insights are crucial for beginning teachers to

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build positive relationships with their students or to restore conflictual relationships and which competencies¹ should thus be prioritized in teacher training. This study addressed this question using the Delphi method. Within this introduction, first, the importance of teacher-student relationships is discussed. Second, challenges and perspectives for teacher education are summarized. Third, relevant theoretical frameworks are presented. Finally, the contributions and aim of the present study are specified.

1. The importance of teacher-student relationships

Teachers build relationships with their class as a group as well as with individual students. Both levels of teacher-student relationships can impact teacher and student well-being (Buyse et al., 2009; Corbin et al., 2019; Nguyen et al., 2020; Rucinski et al., 2018). According to various studies, students who have a close relationship with their teacher tend to have more favorable views toward school, perform better academically, and form more positive relationships with their peers (Ansari, Hofkens et al., 2020; Roorda et al., 2017; Roorda et al., 2020). Conversely, students who have a conflictual relationship with their teacher are at a disadvantage, as they may experience negative cognitive and affective outcomes and the amplification of internalizing and externalizing problems (Ansari, Hofkens et al., 2020; Nguyen et al., 2020; Roorda et al., 2014; Roorda & Koomen, 2021). Notably, students' ability to profit in terms of academic achievements from high-quality classroom environments is conditional upon their dyadic, on-on-one, relationship with the teacher and positive classroom-level relationships cannot compensate for negative dyadic relationships (Buyse et al., 2009; Crosnoe et al., 2010; Nguyen et al., 2020; Rucinski et al., 2018). For instance, high levels of emotional support, classroom organization, and instructional support in the classroom do not improve students' academic performance, unless the student also has a close dyadic relationship with the teacher (Nguyen et al., 2020). Similarly, students who have a conflictual relationship with their teacher are still at risk of underperforming academically, even when emotional support in the classroom is high (Rucinski et al., 2018).

Furthermore, research has shown that conflictual relationships with individual students are a significant cause of stress for teachers, and can lead to burnout symptoms such as emotional exhaustion (Ansari, Pianta et al., 2020; Corbin et al., 2019; Evans et al., 2019; Hagenauer et al., 2015; Spilt et al., 2011). In contrast, close teacher-student relationships can enhance teachers' self-efficacy, sense of personal accomplishment, job satisfaction, and professional motivation (Aboagye et al., 2020; Bosman et al., 2021; Corbin et al., 2019).

2. Building teacher-student relationships and teacher education

Research has established that while teachers consider relationships as vital to their profession (Aspelin et al., 2020), pre-service teachers have difficulties with identifying the teacher-student relationship in classroom situations and rather focus on other classroom elements (e.g. didactic choices; Aspelin & Jonsson, 2019). Moreover, teachers who had recently completed initial teacher education were found to have difficulties applying theoretical concepts related to the teacher-student relationship to their

everyday experiences with children (Koenen et al., 2021). Correspondingly, both researchers (Aspelin & Jonsson, 2019; Blömeke & Kaiser, 2017; Jensen et al., 2015; Jo, 2014; Korpershoek et al., 2016; Rucinski et al., 2018) and policymakers (Directorate-General for Education and Culture, 2010) have argued that teacher education programs might focus too strongly on didactical competencies and not pay sufficient attention to pedagogical, relational and emotional competencies.

It is, however, promising that several intervention studies have shown it is possible to improve teachers' knowledge of teacher-student relationships and provide them with the necessary professional language to discuss these interpersonal aspects of the classroom (e.g. Aspelin et al., 2021; Bosman et al., 2021; Ewe & Aspelin, 2021; Koenen et al., 2021; for a review see: Poling et al., 2022). Shifting teachers' focus from teacher and student characteristics to the interpersonal level can even improve teachers' perceptions of relationship quality as well as their feelings of student-specific self-efficacy (Bosman et al., 2021; Koenen et al., 2021). However, the competencies targeted in existing interventions are either no part of the regular teacher education program or are too scarcely addressed for pre-service teachers to truly understand and apply them to their own teaching practice (Aspelin & Jonsson, 2019; Jensen et al., 2015; Korthagen, 2010).

3. Relationship-building competence

The importance of affective teacher-student relationships has repeatedly been established and several teacher competencies which can contribute to positive, warm relationships have been identified. Importantly, teacher competence has been conceptualized as a multidimensional construct, including teachers' affect-motivation (attitudes or beliefs), (theoretical) knowledge, and situation-specific skills (Baumert & Kunter, 2013; Blömeke & Kaiser, 2017). Each of these aspects separately contributes to teachers' teaching practice (Charalambous, 2015; Depaepe & König, 2018; Horzum & Izci, 2018; Kunter et al., 2013; Spruce & Bol, 2014; Yu, 2018), and the distinct aspects of teachers' relationship-building competence might thus separately contribute to teacher-student relationship quality. Targeting all aspects of competence in initial teacher education, rather than for instance only theoretical knowledge, might strengthen teachers in building positive teacher-student relationships.

Within research on dyadic teacher-student relationships, three frameworks are frequently cited: self-determination theory, attachment theory, and the model of interpersonal teacher behavior (for reviews, see Roorda et al., 2017; Spilt et al., 2022). Each of these frameworks conceptualizes what it means to be competent in building teacher-student relationships and interventions based on different frameworks target different teacher competencies (for reviews, see Kincade et al., 2020; Poling et al., 2022; Spilt et al., 2022). First, within research guided by the self-determination theory, the teacher-student relationship has been identified as an important lever to fulfill both the students' and the teachers' own need for relatedness (Deci et al., 1991; Klassen et al., 2012; Ryan & Deci, 2000). Along with the need for autonomy and the need for competence, every person has three fundamental psychological needs that have to be fulfilled in order to be motivated and to engage meaningfully (Deci et al., 1991; Ryan & Deci, 2000). Teachers can support their students in fulfilling these needs. For instance, teachers can support students' autonomy by offering them choices, can make students feel competent by offering just the right amount of support, or can fulfill students' need for relatedness by lending a listening ear (Opdenakker, 2014). Notably, most intervention studies target students' need for autonomy or competence rather than relatedness (Spilt et al., 2022).

¹ We use the term 'competence' to refer to teachers' affect-motivation (attitudes or beliefs), (theoretical) knowledge, as well as situation-specific skills or strategies, in line with the multidimensional conceptualization of teacher competence by Blömeke and Kaiser (2017).

Second, the attachment framework states that in a positive, effective relationship, characterized by closeness and the absence of conflict, the teacher functions as a 'secure base' and 'safe haven' for students, allowing them to explore the world and supporting their further social, emotional and academic development (Pianta, 1999; Verschueren & Koomen, 2012). Attachment-based interventions emphasize the importance of teachers' sensitivity, reflected in their ability to take the students' perspective as well as to respond appropriately to each student's needs, in building warm, positive relationships (Driscoll & Pianta, 2010; Koomen & Lont, 2004; Pianta, 1999; Vancraeyveldt et al., 2015). Moreover, teachers' reflective functioning, their ability to reflect upon their own and their students' emotions and cognitions, has been identified as a vital skill in developing and restoring relationships (Bosman et al., 2021; Hoogendijk et al., 2019; Slade, 2007; Spilt et al., 2012).

Third, the model of interpersonal teacher behavior describes teacher behavior in the classroom along the dimensions 'Affiliation' and 'Dominance' (Wubbels, Brekelmans et al., 2012; Wubbels et al., 2006; Wubbels, den Brok et al., 2012). Interactions between two parties can be either symmetrical or complementary. Within teacher-student relationships, interactions are often symmetrical with regard to affiliation (i.e. cooperative or friendly behavior from the teacher elicits cooperative or friendly behavior from the student(s)), yet complementary with regard to dominance (i.e. dominant behavior from the teacher elicits submissive behavior from the student(s); Wubbels et al., 2006). However, teachers need to be able to flexibly adapt their behavior to elicit the desired response from the student. When a student for instance shows unkind behavior, a symmetrical reaction from the teacher would be unkind as well. As the teacher aims to elicit friendly behavior, the teacher should instead react unsymmetrically with regard to affiliation and keep initiating positive contact. In the same way, complementary, dominant behavior from the teacher might not be effective in trying to mobilize laid-back students. Directive, dominant behavior would rather reinforce students' passive behavior. In this situation, the teacher would need to act symmetrically, that is by decreasing dominance and directiveness and gradually transferring initiative to the student, in order to change the students' passive behaviors and elicit more active participation (Roorda et al., 2012).

In addition to the competencies emphasized within the frameworks of self-determination theory, attachment theory, and the model of interpersonal teacher behavior, other frameworks propose teacher competencies that might contribute to the quality of their teacher-student relationships. A first competence that might contribute to teacher-student relationships is teachers' general social-emotional competence (self-awareness and self-management). Teachers need to be aware of and be able to cope with their thoughts, emotions, experiences of stress ... in order to retain their sensitivity towards students and be emotionally available to build relationships with their students (e.g. de Ruiter et al., 2021; Jennings, 2015; Koenen et al., 2019). Second, teachers' professional identity, attitudes, and ideas regarding teacher-student relationships might influence how they interact with students. For example, teachers' views on what is an 'appropriate' relationship might impact their relationship-building practices (Kelchtermans, 1996, 2009; O'Connor, 2008). Third, based on the concept of relational competence, teachers' feelings of responsibility as well as their ability to show empathy and respect are expected to contribute to teacher-student relationships (Aspelin et al., 2020, 2021; Aspelin & Jonsson, 2019). Finally, core elements of existing interventions targeting teacher-student relationships might offer insights into contributing teacher competencies (Kincade et al., 2020; Poling et al., 2022). Particularly, direct-proactive practices, such as spending one-on-one time with

students or establishing positive greetings and farewells, might be essential in building positive teacher-student relationships (Kincade et al., 2020).

4. The present study

The assumed lack of attention for relationship-building competencies in teacher education programs might be partly understood, as it might be unclear to researchers, policymakers and teacher educators which competencies should be prioritized in initial teacher education. As discussed, teacher-student relationships have been studied from multiple perspectives, using different conceptualizations and emphasizing different teacher competencies. There is thus a rich academical knowledge base regarding what competencies might contribute to building positive relationships with students. However, it is less clear to what extent current scientific knowledge on relationship-building is important from the perspective of beginning teachers, and should thus be targeted in initial teacher education. This study aimed to address this question of what relational competencies beginning teachers need, including teachers' attitudes, knowledge, and skills. Using the Delphi method, we aimed to obtain a shared point of view from both researchers and teacher educators as to which relationship-building competencies are essential for beginning teachers and need to be prioritized in teacher education. First, a list of competencies was compiled based on the most frequently used theoretical frameworks and interventions. Second, this list was evaluated by an expert panel, to add competencies, select from and prioritize within the list. In this evaluation, we specifically focused on which competencies should be included in initial teacher education programs. Finally, we structured the selected 'need to know' competencies into 5 themes.

5. Method

5.1. Design

The Delphi method was chosen as the most appropriate method to address the study's aim of identifying competencies that are essential for beginning teachers and thus should be part of the initial teacher training program. In a Delphi study, a panel of experts is repeatedly surveyed after which the responses are aggregated and shared with the group. Agreement is expected to increase after each round, with the aim of reaching consensus (Bolger & Wright, 2011). Exchanging views and arguments among a group of experts to reach a shared point of view is preferable over responses from individual experts (Okoli & Pawlowski, 2004; Sitlington & Coetzer, 2015; Skinner et al., 2015). Moreover, the Delphi method was chosen as results from repeated, anonymous surveys are not susceptible to negative effects of group discussions such as participants being influenced by other participants' social or academic standing or the discussion being affected by dominant personalities or group-thinking (Bolger & Wright, 2011; Franklin & Hart, 2007). In this study for instance, teacher educators might perceive their own expert status as less valid compared to that of professors associated with a university and refrain from giving their own opinions.

5.2. Sample

To address the research question, we needed experts who are (1) familiar with the academical knowledge base on relationship-building and (2) with the needs of beginning teachers with regard to relationship-building. We therefore included both postdoctoral-level researchers focused on teacher-student

relationships and lecturers in teacher education in the expert panel. On the one hand, we expected researchers to have advanced knowledge of the scientific literature, yet a less deep understanding of the needs of beginning teachers. On the other hand, we expected lecturers in teacher education to have knowledge of the scientific literature, and a more profound understanding of what knowledge is relevant for beginning teachers. We choose not to include in-service teachers, as they have no academical education and might not be sufficiently familiar with the suggested competencies and concepts and, as a result, could have difficulties judging their importance for teacher training. As we aimed to obtain consensus between both groups of experts, one joint panel was formed. In a Delphi study, the panel size can vary from 4 to 30 panel members (Okoli & Pawlowski, 2004; Skinner et al., 2015). We aimed to include at least 10 panel members, with a balance between lecturers and researchers. However, due to drop-out following the first survey, the final panel consisted of only nine experts. A short description of the expert panel is provided in [Appendix A](#).

Lecturers. Lecturers were recruited from three partner university colleges, participating in the overarching research project. Heads of the pre-primary and primary teacher education programs received a recruiting e-mail, which they forwarded to all lecturers. From each partner university college, at least one lecturer agreed to participate. To retain a balance between lecturers from the different university colleges, as well as a balance between lecturers and researchers, at most two lecturers from each university college were randomly selected for participation in the panel. This resulted in a selection of five lecturers participating in the expert panel.

Researchers. A convenience sample of 10 international researchers, part of the extended professional network of the authors, was contacted via e-mail and invited to participate in the panel. Researchers were chosen based on their expertise in research on teacher-student relationships. The selection method was deemed appropriate, as the goal of a Delphi study is to reach consensus within a group and results thus transcend individual participants. Six researchers, of which one from the authors' university, agreed to participate. However, following the first survey, two researchers indicated their inability to participate in further surveys. These two researchers were not replaced due to time restraints at this point in the study. This resulted in a selection of four postdoctoral-level researchers participating in the expert panel.

5.3. Procedure and materials

Ethical approval was granted prior to the start of the research by the authors' research institute. Before starting the first survey, panel members were presented with the informed consent form and asked to indicate their willingness to participate. If consent was not obtained, the online survey form would be closed and the survey itself not started. All participants agreed and approved the informed consent form.

As we aimed to evaluate the competencies emphasized within existing frameworks in an integrated manner, a first selection of 50 competencies relevant to building teacher-student relationships was compiled based on the literature (see [Appendix B](#)). The aim of a first survey was to complete this list of competencies. Panel members were presented with the compiled list of competencies and asked (1) whether any competencies were missing and (2) whether they had any concerns or questions regarding the list. The competencies were presented in a random order. Following this first survey, 19 suggestions from panel members were added to the list, resulting in a list of 79 competencies. In addition, as some panel members indicated certain items were unclear, a more elaborate description of each competence was added to the list and included as an attachment in the following surveys. In the next step, the

importance of these competencies was repeatedly evaluated and rated by the panel, until agreement was reached or up until a maximum of three rounds ([Meijering et al., 2013](#)).

In the second survey (Rating of importance round 1), the adapted list of 79 competencies was presented to the panel. First, the panel was asked to rate for each competence whether this competence was 'not to know' (i.e. this is not essential for beginning teachers), 'nice to know' (i.e. if there is room in the curriculum, this would be a nice addition) or 'need to know' (i.e. each beginning teacher should know or be able to do this). In Flanders, initial teacher education is offered in a three-year bachelor program. The competencies were rated with this context in mind (i.e. a beginning teacher has completed three years of teacher education). In a follow-up question panel members were asked to give a short argumentation as to why they choose this rating yielding qualitative data. Competencies were again displayed in a random order.

In the third survey (Rating of importance round 2), answers from the previous round were summarized and presented to the panel members. First, an overview was given of those competencies for which consensus was reached (i.e. perfect agreement or at most one rater with an answer deviating at most one scale point) for the ratings of importance (not, nice, or need to know). Competencies by consensus rated as 'nice to know' or 'not to know' were removed from the following questionnaires. Second, those competencies for which consensus was not reached were presented, accompanied by the arguments given by other panel members as to why this should or should not be included in teacher education. The competencies were presented in a random order. Panel members were again asked to rate each competence as 'not to know', 'nice to know', or 'need to know' and give a short argument as to why they choose this rating (yielding qualitative data).

In the fourth survey (Rating of importance round 3), a first question presented the competencies for which consensus was not reached and the arguments given by other panel members as to why they rated 'not', 'nice', or 'need to know'. Panel members were asked to rate the importance of these competencies a final time.

Additionally, in the fourth survey, panel members were presented with a follow-up question targeting those competencies by consensus rated as 'need to know' (as well as those competencies on which no consensus was yet reached). Panel members were asked to indicate what level of mastery beginning teachers should have for each of these competencies, on a scale from 1 (basic) to 3 (advanced). A detailed description of what each level entails for either an attitude, knowledge, or skill was included in the question (see [Table 1](#)). Panel members were able to provide additional comments (yielding qualitative data).

The surveys used in this study included both closed and open-ended questions, resulting in both quantitative and qualitative data. Quantitative data was used to make a selection within the list of competencies (which competencies should be prioritized?). Qualitative data was used to understand the choices of the panel (why should these competencies (not) be prioritized?).

5.4. Data analysis

Following the second, third, and fourth surveys, intraclass correlations and Fleiss' kappa were computed as an overall indication of inter-rater agreement on the ratings of importance ([Cicchetti et al., 2006](#); [Falotico & Quatto, 2014](#); [Fleiss, 1971](#)). For the computation of these indices, items rated as 'not to know' or 'nice to know' were both recoded as 'nice to know', as we were mostly interested in reaching consensus on what competencies are essential and thus 'need to know'. For further analyses (e.g. determination of consensus), the original separate coding was retained.

For the second survey, consensus on individual items was

Table 1
Description of mastery levels.

	Basic level	Intermediate or 'growing' level	Advanced level	Expert level
Attitudes	Describing: getting to know own views, beliefs, ideas, motivations	Interpretation: connecting views, beliefs, motivations to (previous) personal experiences	Critical-analytical: dare to investigate and question your own ideas, views, beliefs and motivations	Creative: use experiences and reflections to develop new ideas, beliefs and motivations
Knowledge	Remember, recognize, describe	Interpret, explain	Analyze, explore connections and relationships	Evaluate, experiment, design and create
Skills	Recognize and mirror in simple, structured contexts/situations	Apply in simple contexts/situations with support	Apply in simple contexts/situations without support	Apply in complex contexts/situations

Note: We did not take into account the "expert level", as we expect teachers to only reach this level on the job (in the context of lifelong learning; following diverse experiences and reflections in diverse situations and contexts), and not during initial teacher training.

operationalized as either full agreement between panel members (i.e. all panel members gave the same rating) or majority agreement between all but one panel members. In the case of majority agreement, the divergent rating deviated at most one scale point from the other raters (i.e. if all but one rater indicated 'need to know', the deviating rater indicated 'nice to know'; if all but one rater indicated 'not to know', the deviating rater indicated 'nice to know'; if all but one rater indicated 'nice to know', the deviating rater indicated either 'need to know' or 'not to know'). For the third and fourth surveys, consensus on individual items was again operationalized as either full agreement between panel members (i.e. all panel members gave the same rating) or majority agreement. For these surveys, majority was defined as at least two thirds of the panel (i.e. at most three deviating ratings) with the divergent ratings deviating at most one scale point from the other ratings.

Finally, once all competencies were rated, all qualitative data (i.e., written argumentations) were compiled for each item. These written arguments were all thoroughly read to gain insight into the justifications provided by the panel for their 'need to know', 'nice to know', or 'not to know' ratings.

6. Results

6.1. Ratings of importance and selection of competencies

Following the first survey, 19 suggestions from panel members were added to the list (indicated with an asterisk in [Appendix B](#)), resulting in a total of 79 items to be rated in the following surveys.

For the second survey, intraclass correlations ($ICC = 0.28$ [0.21; 0.36], $F(78,468) = 3.70$, $p < .001$) and Fleiss' κ ($\kappa = 0.20$, $Z = 8.27$, $p < .001$) indicated only poor to slight overall agreement between panel members on the ratings of importance. However, consensus was reached for 37 items (46.8% of the 79 items; 18 times full agreement, 19 times majority agreement). Thirty competencies (38.0%) were rated as 'need to know'. Seven competencies (8.7%) were rated as 'nice to know' and removed from further surveys. No competencies were rated as 'not to know'.

For the third survey, intraclass correlations ($ICC = 0.47$ [0.39; 0.55], $F(78,624) = 8.80$, $p < .001$) and Fleiss' κ ($\kappa = 0.47$, $Z = 25.30$, $p < .001$) increased and indicated slight to fair overall agreement on the ratings of importance. Consensus was reached on an additional 34 items (43.0%; all majority agreement), resulting in consensus on 71 items (89.9%) in total. Two items (2.5%) were rated as 'need to know', the remaining 32 items (40.5%) were rated as 'nice to know' and removed from further surveys. No competencies were rated as 'not to know'.

For the fourth survey, intraclass correlations ($ICC = 0.47$ [0.41; 0.57], $F(78, 624) = 9.30$, $p < .001$) and Fleiss' κ ($\kappa = 0.49$, $Z = 26.12$, $p < .001$) only slightly increased and indicated fair to moderate

overall agreement on the ratings of importance. Consensus was reached on five of the remaining eight items (all majority agreement, four rated 'need to know', one rated 'nice to know', no competencies rated 'not to know'), resulting in consensus on 76 items (96.2%) in total. On three items, the panel remained divided following the final round of ratings. As no agreement was reached on whether these competencies were 'need to know', it was concluded by the researchers that these should be considered 'nice to know'. In total, 43 items were by consensus rated 'nice to know' and 36 items were rated 'need to know'. A visual representation of the subsequent survey results is provided in [Fig. 1](#).

An overview of all competencies, importance ratings by the panel, and the final decision on importance is provided in [Appendix C](#). The selected competencies can be sorted into five categories which are presented in separate tables ([Tables 2A](#) through [2E](#)): (1) The importance of the teacher-student relationship for students, (2) Key concepts to understand and describe teacher-student relationships, (3) Attitudes and reflective functioning, (4) Specific strategies to establish and maintain relationships, and (5) Specialization: building relationships in the context of student diversity. A more detailed description of these results and the qualitative analysis of the panel's arguments to substantiate and interpret the quantitative results is provided in the discussion.

6.2. Ratings of mastery level

We received a response from all panel members on all surveys. However, in the final survey four panel members left the additional mastery level question incomplete and rather gave a more elaborate description of their ideas in a comment or separate e-mail to the researcher. The mastery level indications and accompanying remarks thus served mainly as a (qualitative) inspiration to prioritize within the competencies rated as 'need to know'. The mean rated level of mastery, expected to be attained at the end of initial three-year teacher education, was included in [Tables 2A](#) through [2E](#). For four competencies, the panel expected only a basic level of mastery. For 21 competencies an intermediate level and for 11 competencies an advanced level was desired. For the competencies related to the importance of the teacher-student relationship and key concepts (i.e. mostly theoretical knowledge, [Table 2A](#) and [Table 2B](#)), at most an intermediate level was expected. With regard to the attitudes (empathy, respect, non-prejudice) and reflective functioning competencies (self-awareness, professional identity, teachers' responsibility), an advanced or intermediate level was expected, respectively ([Table 2C](#)). For several of the specific strategies ([Table 2D](#)) an advanced mastery level was expected (e.g. positive greetings and farewells), while for others an intermediate level was deemed sufficient (e.g. perspective taking). For competencies related to student diversity a basic level was desired ([Table 2E](#)).

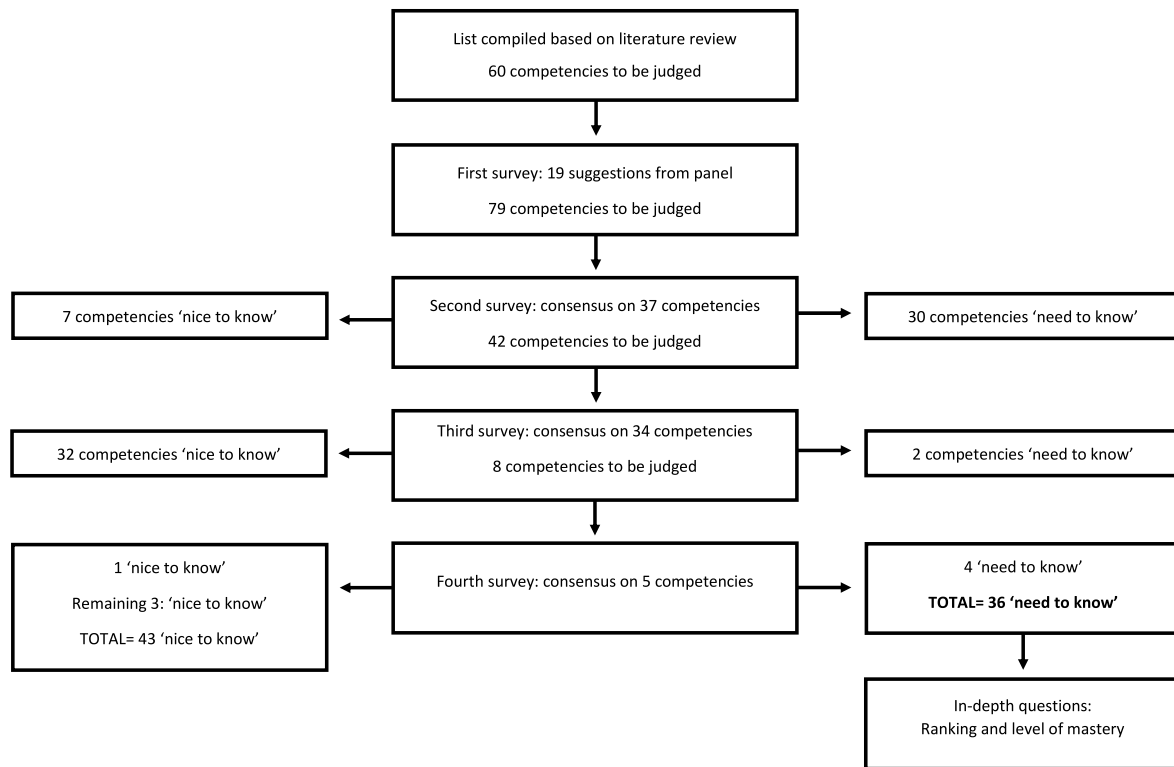


Fig. 1. Flowchart of subsequent survey results.

Table 2A

Selected competencies: Importance of the teacher-student relationship.

	Short description	Level of mastery (see Table 1)
1	The importance of teacher-student relationships for the personal well-being of the student	Intermediate
2	The importance of teacher-student relationships for the social-emotional-motivational development of the student	Intermediate
3	The importance of teacher-student relationships for the cognitive development of the student	Intermediate
4	The importance of teacher-student relationships for the development of the student's relationships with classmates	Intermediate

Table 2B

Selected competencies: Key concepts.

	Short description	Level of mastery (see Table 1)
24	Attachment theory: Central concepts such as secure base and safe haven and autonomous exploration	Intermediate
27	Attachment theory: Describing relationship quality in terms of closeness, conflict and dependency	Intermediate
28	Attachment theory: Teachers' mental representations or internal working models of teacher-student relationships	Basic
30	Attachment theory: Teacher sensitivity	Intermediate
31	Self-determination theory: Importance of teacher-student relationship from a motivational perspective	Intermediate
32	Self-determination theory: Need for Autonomy, Belonging and Competence (ABC)	Intermediate
33	Self-determination theory: Supporting students' ABC needs via autonomy support, involvement, structure	Intermediate

Table 2C

Selected competencies: Attitudes and reflective functioning.

	Short description	Level of mastery (see Table 1)
66	Showing empathy	Advanced
67	Showing respect	Advanced
77	Non-prejudice	Advanced
42	Self-awareness: Ideas and thoughts about individual students (e.g. attributions of problem behavior)	Intermediate
51	Professional identity regarding building relationships: professional dimension (e.g. what is an 'appropriate' relationship?)	Intermediate
53	Ideas or beliefs about the teachers' responsibility for the teacher-student relationship	Intermediate

Table 2D
Selected competencies: Specific strategies.

	Short description	Level of mastery (see Table 1)
54	Perspective taking	Intermediate
55	The student's emotion and stress regulation	Intermediate
56	Praise	Advanced
58	Positive to negative ratio of interactions	Advanced
59	One-on-one time	Intermediate
61	Positive greetings and farewells	Advanced
62	Validating and labeling the student's emotions and views	Intermediate
64	Non-verbal behavior (e.g. gestures)	Advanced
65	Posture (e.g. get down to your knees)	Intermediate
69	Observation skills	Advanced
71	Peer-conflict resolution	Intermediate
72	Sensitive behavior management	Intermediate
73	Offering help (proactive and reactive)	Advanced
74	Showing interest in the student (their interests, daily life ...)	Advanced
75	Express positive expectations	Advanced
76	Non-violent communication	Intermediate
79	Competences related to parents	Basic

Table 2E
Selected competences: Student diversity.

	Short description	Level of mastery (see Table 1)
15	Building teacher-student relationships with students with special educational needs	Basic
21	Building teacher-student relationships with students from different (social-economic, cultural, religious ...) backgrounds	Basic

7. Discussion

The aim of this Delphi study was to identify those competencies that pre-service teachers should obtain during initial three-year teacher education to be prepared to build positive, warm relationships with individual students. An expert panel consisting of lecturers in teacher education and academic researchers selected 36 'need to know' competencies. In order to structure the discussion, we present the results in 5 categories: (1) the importance of teacher-student relationships for students, (2) key concepts to understand and describe teacher-student relationships, (3) attitudes and reflective functioning, (4) specific strategies to establish and maintain relationships, and (5) building relationships in the context of student diversity. In the following paragraphs, the selected 'need to know' competencies will be presented, accompanied by the argumentations given by the panel as to why competencies were or were not chosen. Additionally, connections with relevant literature will be explored.

7.1. Importance of the teacher-student relationship (Table 2A)

First, the panel indicated that a critical step in building relationship-building competence is to motivate teachers and create awareness of the importance of positive, warm teacher-student relationships. The panel selected four items contributing to this goal and concluded that teachers need to be aware of the importance of teacher-student relationships for the personal well-being of the student, the social-emotional-motivational development of the student, the cognitive development of the student as well as for the development of the student's relationships with classmates.

Notably, no items related to the importance of teacher-student relationships for the teacher were selected. The panel members felt the focus on the benefits of the teacher-student relationship for the student would be more motivating for (preservice) teachers compared to a focus on the teacher. Items related to the importance of teacher-student relationships for the professional and personal well-being of the teacher were rated 'nice to know', and could thus

be added if there is room in the curriculum (see Appendix C). It is important to note that on item 7 *'the importance of teacher-student relationships for the professional well-being of the teacher'* no consensus was reached. Five out of nine panel members did consider this a 'need to know' competence. Panel members indicated that they selected those items which are most at play and visible in the classroom itself, rather than more indirect effects of teacher-student relationships. For instance, they selected item 4 *'the importance of teacher-student relationships for the development of the student's relationships with classmates'* as 'need to know', yet item 5 *'the importance of teacher-student relationships for the development of the student's relationships with peers'* was rated as 'nice to know'. The relevance of teacher-student relationships for the development of peer relationships has been established with research indicating that close teacher-student relationships are beneficial for peer acceptance, while conflictual teacher-student relationships may hinder peer relationships (Endedijk et al., 2022; ten Bokkel et al., 2022).

7.2. Key concepts (Table 2B)

Second, to build positive teacher-student relationships, teachers require a basic conceptual understanding of what these relationships look like and how they can be understood. The panel decided that teachers should have a basic understanding of central concepts from attachment theory (e.g. secure base, safe haven) and self-determination theory (e.g. need to belong). All items related to the theory of interpersonal teacher behavior were considered 'nice to know'. This selection of theories is consistent with the literature, as attachment theory and self-determination theory are the most frequently used in research on dyadic teacher-student relationships (Roorda et al., 2017; Spilt et al., 2022), whereas the theory of interpersonal behavior is most often applied at the classroom level (Wubbels, Brekelmans et al., 2012; Wubbels et al., 2006; Wubbels, den Brok et al., 2012).

The panel emphasized that a thorough, theoretical understanding of these theories is not required. Basic knowledge of the central concepts applied to the educational setting is sufficient. The

panel raised the idea of “giving language”: this knowledge of attachment and self-determination theory is crucial, as teachers need a shared language to describe and understand the relationships they build with their students. This idea is in line with recent research that showed teachers seem to lack the language to discuss the relational aspects of the classroom (Aspelin et al., 2021; Koenen et al., 2022). In addition, it can be challenging for (beginning) teachers to apply theoretical concepts to their own relational experiences in the classroom (Koenen et al., 2021).

Teacher training should thus provide teachers with a basic level of theoretical knowledge and encourage them to use this knowledge to discuss their own relational experiences in the classroom.

7.3. Attitudes and reflective functioning (Table 2C)

Third, the panel indicated that a positive attitude of showing empathy, respect, and no prejudice towards students is essential in building positive relationships with them. Related, three items about the teacher's self-awareness and professional identity were selected as ‘need to know’. Teachers should be aware of and reflect on their ideas and thoughts about individual students (e.g. Why is this student showing disruptive behavior?), their ideas or beliefs about their responsibility for the teacher-student relationship (e.g. Should I or should the student take the first step following a conflict?), and the professional dimension of their professional identity with regard to teacher-student relationships (e.g. What do I want my relationships with students to look like?). Other dimensions (moral, emotional, and performative) were considered ‘nice to know’. The selected competencies show several similarities with the conceptualization of relational competence in other educational studies, including the importance of teachers' responsibility and qualities such as empathy and respect (Aspelin et al., 2020, 2021; Aspelin & Jonsson, 2019; Koenen et al., 2022; McGrath & Van Bergen, 2019).

7.4. Specific strategies (Table 2D)

Fourth, the panel selected several specific strategies which can be used to improve interactions between teachers and students and in that way improve their relationship. Most of these strategies (e.g. one-on-one time, praise, validating and labeling the student's emotions) have been identified as crucial elements of interventions targeting teacher-student relationships from an attachment perspective (Cook et al., 2019; Driscoll et al., 2011; Kincade et al., 2020; Vancraeyveldt et al., 2015). Panel members highlighted the importance of these specific strategies, especially for preservice teachers. They suggest asking preservice teachers to apply these strategies is an ideal way to increase their awareness of and attention to individual teacher-student relationships and help them transfer theory to practice (Korthagen, 2010).

Notably, the panel rated the item ‘competencies related to parents’ as ‘need to know’. This is a competency that is not directly applicable in the classroom itself and seems therefore in contrast with the before stated motivation to choose items exactly because they are closely related to the classroom practice. However, in the rankings and additional remarks, panel members did indicate that these parent-related competencies are of less importance compared to other ‘need to know’ competencies. As research has demonstrated a link between parent-teacher and teacher-student relationships and the interplay of both can impact students' socio-emotional development, it might be worthwhile to include these competencies (Hughes & Kwok, 2007; Jeon et al., 2021; Lang et al., 2020).

7.5. Specialization: student diversity (Table 2E)

Finally, two items related to building relationships with diverse groups of students were selected as ‘need to know’, yet the panel indicated that they were of less importance compared to other competencies. Some panel members concluded that the teacher-student relationship should be discussed when other aspects related to diversity are discussed, while other panel members suggested these competencies should be addressed in additional training (e.g. special educational needs training) rather than initial teacher education.

As students with for instance special educational needs are more likely to develop negative, conflictual relationships with their teachers, it might be beneficial to alert teachers to this risk (McGrath & Van Bergen, 2015). If teachers are more aware of which students are at-risk, they might be motivated to invest more time and energy in building relationships with these students. However, the panel also raised a concern for self-fulfilling prophecy if there is too big an emphasis on at-risk groups. The panel indicated that without offering concrete tools to overcome difficulties in building relationships with at-risk students, teachers might simply point to student characteristics as the cause of their negative relationship and forgo their responsibility to try and restore the relationship. Indeed, if teachers are prone to attribute problems to stable – child or family – characteristics, they might stop trying to improve the relationship as they “expect no change to result from their efforts” (Koenen et al., 2022, p. 8).

7.6. Competencies rated ‘nice to know’

Three categories of competencies were mainly rated ‘nice to know’, namely items related to (1) child, teacher, and school characteristics, (2) teacher self-awareness and self-management, and (3) professional identity. In the following paragraphs, the three categories are discussed. Motivations of the panel as to why these competencies were not deemed ‘need to know’ are explored and related to previous research.

First, competencies regarding student, teacher, and school characteristics that are of interest to the teacher-student relationship (e.g. item 11 ‘Student characteristics related to the (quality of the) teacher-student relationship’) were rated ‘nice to know’ rather than ‘need to know’. Panel members indicated several reasons for their caution in including this topic in teacher education. For instance, they were worried a focus on student characteristics might result in self-fulfilling prophecies. As teacher expectations with regard to students' achievement can influence students' actual achievement (e.g. Gentrup et al., 2020), the concern that teacher expectations with regard to the teacher-student relationship can influence the actual relationship quality might be justified. It is however unclear whether this knowledge would lead to a self-fulfilling prophecy (“I will probably not be able to build a warm relationship with this student, so why should I invest my time?”) or rather help teachers to identify those children which would benefit from a close relationship (“I might be harder to build a warm relationship with this student, so I will have to put in more of an effort”). The panel members seem to err on the side of caution and do not prioritize this knowledge. With regard to teacher and school characteristics, panel members referred to the fact that research on these topics is still scarce, and often inconsistent. Therefore, they opted not to include these competencies. Moreover, panel members again evaluated competencies on their practical relevance, stating for instance: “We cannot change these student characteristics. So is it even useful to include them?” and “This is more of theoretical interest than practical relevance”.

Second, only one competence related to teacher self-awareness was selected as 'need to know', namely the item *'Ideas and thoughts about individual students'*. Other competencies related to self-awareness and all competencies related to self-management were rated as 'nice to know' and for the item *'Emotions experienced by teachers in relationships with students'* no consensus was reached. This is in contrast with research emphasizing the importance of teacher self-care for building and restoring teacher-student relationships, teacher well-being, and job satisfaction as well as teaching practice (e.g. Beltman et al., 2011; Brown et al., 2017; Chang, 2009; de Ruiter et al., 2021; Hastings & Brown, 2002; Jennings, 2015). Several interventions aimed at improving teacher-student relationships for instance first aim to increase the teacher's awareness of their feelings and thoughts in individual relationships before trying to elicit change (Bosman et al., 2021; Hoogendijk et al., 2019; Koenen et al., 2021). Moreover, self-awareness and self-management can be critical in remaining resilient in the face of challenges, such as difficulties establishing a positive relationship with students (Beltman et al., 2011; Hastings & Brown, 2002). Notably, building teacher resilience within teacher education programs has recently received more attention both internationally (Mansfield, 2020) and within the Flemish context, with several projects aiming to increase pre-service teachers' resilience (e.g. Hens, 2021; Vanderhaeghen, 2021). It is noteworthy that the decision of the panel does not align with these recent developments. One possible explanation is that the panel was strictly focused on relationship-building competencies that need to be addressed in initial three-year teacher education. Teacher self-awareness and self-management might mainly develop on the job, when establishing long-lasting relationships with students. The panel might feel that these competencies are thus more suited to be included in advanced education or professional development initiatives for in-service teachers and thus not select them as 'need to know' for initial teacher education. In their argumentations the panel did link self-awareness and self-management to the students' experiences in practice, stating for instance: *"This [self-awareness of one's emotions] wouldn't be a priority for me in the teacher education program as a whole. But we should pay attention to this during internship meetings indeed."* and *"Students indicate that they consider the step from the [teacher education] program to the professional field to be a big one. Making them aware of their own emotions as a teacher can, in my opinion, be a good step to (partially) prevent the outflow from the teaching profession."*

Third, as discussed, only the professional dimension of professional identity (e.g. what is an 'appropriate' relationship?) was selected as 'need to know', whereas the moral, emotional and performative dimensions were rated 'nice to know'. Research has shown that teachers regard their relationships with students as central to their work, yet they experience challenges in building and understanding these relationships (Aspelin et al., 2020; García-Moya et al., 2019; Koenen et al., 2021; O'Connor, 2008). We therefore argue that the importance of teacher-student relationships for the development of teachers' professional identity should not be overlooked. Importantly, the panel prioritized within several dimensions of professional identity, yet seemed to prefer a more integrative approach for the discussion of teachers' professional identity and the importance of teacher-student relationships for its development. Panel members indicated they selected within the different aspects of professional identity, judging the professional dimension to be more important compared to other aspects. For instance, panel members noted: *"Being aware of your professional identity is very important, but the moral dimension felt to me more like an extra addition, compared to the professional dimension, and I thus rated this 'nice to know'."* and *"I just made choices within the identity dimensions, this could definitely be relevant as well."* Additionally,

panel members argued to use an integrative approach and allow discussion of all dimensions of professional identity.

8. Limitations

The Delphi method is a well-established method to obtain consensus among experts on a particular topic and has several strengths (Bolger & Wright, 2011; Humphrey-Murto & de Wit, 2019). However, some limitations, inherent to the method itself or a result of our choices, need to be taken into account as well. First, the Delphi method is especially valuable to create a discussion among people with different backgrounds and professions (e.g. researchers versus lecturers) and move towards consensus (Bolger & Wright, 2011). The panel members in this study quickly agreed on the importance of a selection of competencies, resulting in only minor debates. The absence of strong discussions might have caused us to miss out on detailed argumentations or ideas. Second, in this study we formed a diverse panel to include the perspective of both researchers and lecturers in teacher education, both having extensive knowledge of the scientific literature but with different interests (i.e., academic research and education vs. professional education). In this study, in-service teachers, who typically have no academic education and cannot be expected to know the academic literature were not included. However, future research could explore whether in-service teachers would agree with the decisions of this panel. Third, the final panel consisted of researchers from three countries and lecturers in teacher education only from Flanders. Answers from the panel might thus be influenced by the Flemish school context and might not be directly applicable to other cultures or school systems. Moreover, the panel rated which competencies should be obtained following a three-year bachelor program, as this is how initial teacher education is offered in Flanders. Other educational systems (e.g. a four-year program) might have the opportunity to focus on additional competencies or develop a deeper level of mastery. Fourth, we immediately presented the panel members with a list of competencies, based on the literature. This might have 'primed' the panel to think in a certain direction. Finally, once consensus was reached on the importance of a certain competence, the importance of this competence was not questioned again. A different answering format (e.g. 'drag and drop' all competencies into 'not', 'nice' or 'need to know'- categories instead of a three-point rating scale for each competence) or presenting the panel members with an overview of their own ratings before submitting (e.g. in a visual representation of the three-point rating categories) might have stimulated additional reflection and resulted in a different selection of 'need to know' competencies.

9. Recommendations for research and teacher education

The results of this study can inspire future research. It might for instance be interesting to investigate whether and how the selected 'need to know' competencies are already included in current teacher education curricula. As lecturers in teacher education programs were involved in the selection process, results of the study might partly reflect the current curricula. In addition, the results of the study might be replicated using a different research method (e.g. individual or focus group interviews) or a different sample (e.g. pre- and in-service teachers) as more research is needed on how academic knowledge on teacher-student relationships is relevant for practice. Moreover, results of this study can inspire teacher training. The selected competencies can be rephrased as learning goals and inform the development of targeted learning materials and activities to increase teachers' relationship-building competence. For the aim of this study, the competencies were described in a very specific way, targeting

distinct knowledge, skills, and attitudes. Thus, several overarching theories or concepts were divided into separate items to be able to distinguish and prioritize within larger competencies. As argued by our panel members, a more integrative approach, combining several items into one competence-goal, will be necessary in translating the list of competencies to practice.

Guided by the multidimensional conceptualization of teacher competence (including teachers' affect-motivation (attitudes or beliefs), (theoretical) knowledge, and situation-specific skills; Baumert & Kunter, 2013; Blömeke & Kaiser, 2017), diverse learning activities and materials would have to be developed, as each of the aspects of competence separately contribute to teachers' teaching practice (Charalambous, 2015; Depaepe & König, 2018; Horzum & Izci, 2018; Kunter et al., 2013; Spruce & Bol, 2014; Yu, 2018). For example, to develop pre-service teachers' knowledge (e.g. knowing that self-determination theory distinguishes three basic needs) repetition, rehearsal and review strategies can be applicable, while for the development of situation-specific skills (e.g. giving praise, perspective taking) modeling and overt practice might be more appropriate (Morrison et al., 2019). Additionally, offering a variety of learning activities can help to keep students active and motivated (Owen et al., 2008). Furthermore, recent research has shown that it can be difficult for (beginning) teachers to apply theoretical knowledge to their teaching practice (Aspelin et al., 2021; Koenen et al., 2022). The benefits of providing sufficient opportunities throughout the years to apply competencies learned during teacher training in practice cannot be underestimated.

Finally, the rated mastery levels can help to prioritize within the list of competencies. For instance, competencies for which an intermediate or advanced level was desired (e.g. observation skills) should be sufficiently included and developed during initial teacher education. Competencies for which a basic level was deemed sufficient (e.g. knowledge of the concept of mental representations) can be introduced during initial teacher education, yet further developed in specialized training or on the job, with the support of professional development initiatives.

10. Conclusion

Reports from beginning teachers indicate that they have difficulties building teacher-student relationships. Although there is a rich academic knowledge base, it is unclear which scientific insights are particularly valuable for beginning teachers and should thus be targeted in initial teacher education to prepare beginning teachers to build positive, warm relationships with their students. Using the Delphi method, this study aimed to provide a guideline for teacher educators for the selection of essential 'need to know' competencies to be addressed in their educational programs as well as to provide teacher educators with an idea of at which level of

mastery these competencies should be taught. Thirty-six competencies, related to the importance of teacher-student relationships for students, key concepts to understand and describe relationships, attitudes and reflective functioning, specific strategies to establish and maintain relationships, and building relationships in the context of student diversity, were selected as 'need to know' by an expert panel consisting of lecturers in teacher education and academic researchers. The panel argued to select mostly competence items from the survey that are closely related to what happens in the classroom, can easily be translated to practice, and therefore have a clear, practical relevance for the teacher. For the majority of competencies, at least an intermediate level of mastery was expected at the end of initial teacher training. For four competencies, the panel expected only a basic level of mastery at the end of initial education. For the majority of competencies, an intermediate (21 competencies) or advanced level (11 competencies, mainly related to attitudes and specific strategies) was desired. Results can inspire teacher education programs to critically reflect on the contents of their curriculum.

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Author contributions

LB: Conceptualization; Methodology; Investigation; Formal analysis; Project administration; Funding acquisition; Writing – original draft, review & editing. JS: Conceptualization; Supervision; Writing – review & editing.

Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Data availability

The data that has been used is confidential.

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Appendix A

Description of panel members

Panel member	Affiliation	Job description
1	University college A (Belgium)	Lecturer in teacher education (pre-primary and primary education; 8 years of experience)
2	University college A (Belgium)	Involved in (practice-oriented) research projects
3	University college B (Belgium)	Head of teacher education program (1 year experience) and lecturer (pre-primary and primary education; 8 years of experience)
4	University college B (Belgium)	Involved in (practice-oriented) research projects
5	University college C (Belgium)	Lecturer in teacher education (primary education; 7 years of experience)
		Involved in (practice-oriented) research projects
		Lecturer in teacher education (pre-primary and primary education; 8 years of experience)
		Lecturer (pre-primary education; over 10 years of experience)
		Involved in (practice-oriented) research projects

Appendix A (continued)

Panel member	Affiliation	Job description
6	University A (Belgium) (note: university of the authors)	Researcher; over 20 years of experience as principal investigator
7	University B (The Netherlands)	Researcher; 7 years of experience as principal investigator Also involved in the University's academic teacher education program
8	University B (The Netherlands)	Researcher; over 20 years of experience as principal investigator
9	University C (Germany)	Researcher; over 10 years of experience as principal investigator
—	University D Virginia, USA	Researcher; over 15 years of experience as a principal investigator Note: dropped out following first survey
—	University E Minnesota, USA	Researcher; 2 years of experience as postdoctoral researcher Note: dropped out following first survey

Appendix B

Compilation of relevant competencies

The table below gives an overview of the selected competencies as well as the corresponding theoretical framework or source and a non-exhaustive list of references. Competencies added by the panel members in a later stage of the study are indicated with an asterisk (*).

	Short description	Source	Example references
1	The importance of teacher-student relationships for the personal well-being of the student	Derived from several theoretical perspectives and empirical research	(McGrath & Van Bergen, 2015; Roorda et al., 2017, 2020.)
2	The importance of teacher-student relationships for the social-emotional-motivational development of the student	Derived from several theoretical perspectives and empirical research	(McGrath & Van Bergen, 2015; Roorda et al., 2017, 2020; Weyns et al., 2019)
3	The importance of teacher-student relationships for the cognitive development of the student	Derived from several theoretical perspectives and empirical research	(Engels et al., 2021; McGrath & Van Bergen, 2015)
4	The importance of teacher-student relationships for the development of the student's relationships with classmates	Derived from several theoretical perspectives and empirical research	(Endedijk et al., 2022; Koenen et al., 2022; McGrath & Van Bergen, 2015; ten Bokkel et al., 2022; Verschueren, 2015)
5*	The importance of teacher-student relationships for the development of the student's relationships with peers	Derived from several theoretical perspectives and empirical research	(Endedijk et al., 2022; Koenen et al., 2022; McGrath & Van Bergen, 2015; ten Bokkel et al., 2022; Verschueren, 2015)
6*	The importance of teacher-student relationships for the development of the student's relationships with other adults	Derived from several theoretical perspectives and empirical research	(McGrath & Van Bergen, 2015; Sabol & Pianta, 2012)
7	The importance of teacher-student relationships for the professional well-being of the teacher	Derived from several theoretical perspectives and empirical research	(Ansari, Pianta et al., 2020; Corbin et al., 2019; Hagenauer et al., 2015; Spilt et al., 2011)
8	The importance of teacher-student relationships for the personal well-being of the teacher	Derived from several theoretical perspectives and empirical research	(Ansari, Pianta et al., 2020; Corbin et al., 2019; Hagenauer et al., 2015; Spilt et al., 2011)
9	The importance of teacher-student relationships for the teacher's self-efficacy	Derived from several theoretical perspectives and empirical research	(de Jong et al., 2014; Zee et al., 2017; Zee & Koomen, 2016)
10*	Cultural differences and similarities in teacher-student relationships	Derived from several theoretical perspectives and empirical research	(Chen et al., 2019; Chen, Lindo et al., 2021; McGrath & Van Bergen, 2015)
11	Student characteristics related to the (quality of the) teacher-student relationship	Derived from several theoretical perspectives and empirical research	(Choi & Dobbs-Oates, 2015; McGrath & Van Bergen, 2015; Pianta et al., 2003)
12	Characteristics of the student's environment (e.g. home environment) related to the (quality of the) student-teacher relationship	Derived from several theoretical perspectives and empirical research	(Choi & Dobbs-Oates, 2015; Jeon et al., 2021; Lang et al., 2020; McCormick et al., 2017)
13	Teacher characteristics related to the (quality of the) teacher-student relationship	Derived from several theoretical perspectives and empirical research	(Castle et al., 2015; Choi & Dobbs-Oates, 2015; de Jong et al., 2014; McGrath & Van Bergen, 2019; Yoon, 2002)
14	Characteristics of the school or class related to the (quality of the) teacher-student relationship	Derived from several theoretical perspectives and empirical research	(Choi & Dobbs-Oates, 2015; McGrath & Van Bergen, 2015; Trang & Hansen, 2020)
15*	Building relationships with students with special educational needs	Derived from several theoretical perspectives and empirical research	(McGrath & Van Bergen, 2015; Murray & Greenberg, 2001; Nurmi, 2012; Zee et al., 2020)
16	Building relationships with students with a physical disability	Derived from several theoretical perspectives and empirical research	(McGrath & Van Bergen, 2015; Nurmi, 2012)
17*	Building relationships with students with a (mild) mental disability	Derived from several theoretical perspectives and empirical research	(McGrath & Van Bergen, 2015; Nurmi, 2012)

(continued on next page)

Appendix B (continued)

Short description	Source	Example references
18 Building relationships with students with learning problems or disorders	Derived from several theoretical perspectives and empirical research	(McGrath & Van Bergen, 2015; Nurmi, 2012)
19 Building relationships with students with behavioral or emotional problems or disorders	Derived from several theoretical perspectives and empirical research	(McGrath & Van Bergen, 2015; Nurmi, 2012; Zee et al., 2020; Zendarski et al., 2020)
20 Building relationships with students with attachment problems or disorders	Derived from several theoretical perspectives and empirical research	(Koenen et al., 2019; McGrath & Van Bergen, 2015; Vervoort et al., 2014)
21* Building relationships with students from different (social-economic, cultural, religious ...) backgrounds	Derived from several theoretical perspectives and empirical research	(Chen et al., 2019; Chen, Lindo et al., 2021; McGrath & Van Bergen, 2015; Zendarski et al., 2020)
22 How the teacher-student relationship relates to other concepts (e.g. class climate, behavior management)	Derived from several theoretical perspectives and empirical research	(Buyse et al., 2008; Doyle et al., 2022; Nguyen et al., 2020; Rucinski et al., 2018)
23 Attachment theory: Background of attachment theory (e.g. Bowlby, Harlow, Ainsworth)	Extended attachment perspective	(Pianta, 1999; Roorda et al., 2017; Sabol & Pianta, 2012; Verschueren & Koomen, 2012)
24 Attachment theory: Central concepts such as secure base and safe haven and autonomous exploration	Extended attachment perspective	(Pianta, 1999; Roorda et al., 2017; Sabol & Pianta, 2012; Verschueren & Koomen, 2012)
25 Attachment theory: Similarities and differences between parent-child and teacher-student relationships	Extended attachment perspective	(Pianta, 1999; Roorda et al., 2017; Sabol & Pianta, 2012; Verschueren & Koomen, 2012)
26 Attachment theory: Role of the teacher as an ad hoc attachment figure	Extended attachment perspective	(Pianta, 1999; Roorda et al., 2017; Sabol & Pianta, 2012; Verschueren & Koomen, 2012)
27 Attachment theory: Describing relationship quality in terms of closeness, conflict and dependency	Extended attachment perspective	(Pianta, 1999; Roorda et al., 2017; Sabol & Pianta, 2012; Verschueren & Koomen, 2012)
28 Attachment theory: Teachers' mental representations or internal working models of teacher-student relationships *	Extended attachment perspective	(Koenen et al., 2019; Pianta, 1999; Roorda et al., 2017; Sabol & Pianta, 2012; Verschueren & Koomen, 2012)
29 Attachment theory: The student's mental representations or internal working models *	Extended attachment perspective	(Pianta, 1999; Roorda et al., 2017; Sabol & Pianta, 2012; Verschueren & Koomen, 2012)
30 Attachment theory: Teacher sensitivity	Extended attachment perspective	(Koenen et al., 2019; Pianta, 1999; Roorda et al., 2017; Sabol & Pianta, 2012; Verschueren & Koomen, 2012)
31 Self-determination theory: Importance of teacher-student relationship from a motivational perspective	Self-determination theory	(Deci et al., 1991; Opdenakker, 2014; Roorda et al., 2017; Ryan & Deci, 2000)
32 Self-determination theory: Need for Autonomy, Belonging and Competence (ABC)	Self-determination theory	(Deci et al., 1991; Opdenakker, 2014; Roorda et al., 2017; Ryan & Deci, 2000)
33 Self-determination theory: Supporting students' ABC needs via autonomy support, involvement, structure	Self-determination theory	(Deci et al., 1991; Opdenakker, 2014; Roorda et al., 2017; Ryan & Deci, 2000)
34 Theory of Interpersonal Teacher Behavior: Theoretical background (systems communication theory, Leary Model, Wubbels & Brekelmans)	Theory of interpersonal teacher behavior	(Roorda et al., 2012; Wubbels et al., 2006; Wubbels, den Brok et al., 2012)
35 Theory of Interpersonal Teacher Behavior: Describing interactions between partners based on two dimensions Dominance (dependent versus dominant) and Affiliation (hostile versus friendly)	Theory of interpersonal teacher behavior	(Roorda et al., 2012; Wubbels et al., 2006; Wubbels, den Brok et al., 2012)
36* Theory of Interpersonal Teacher Behavior: Interpersonal circle and 8 possible positions	Theory of interpersonal teacher behavior	(Roorda et al., 2012; Wubbels et al., 2006; Wubbels, den Brok et al., 2012)
37 Theory of Interpersonal Teacher Behavior: Teachers' behavior elicits complementary behavior from the student and vice versa ((a) symmetry and (non)complementarity in student-teacher interactions)	Theory of interpersonal teacher behavior	(Roorda et al., 2012; Wubbels et al., 2006; Wubbels, den Brok et al., 2012)
38 Self-awareness: Emotions experienced by beginning teachers in relation to their tasks as a teacher (e.g. helplessness)	Social-emotional competence Core elements of interventions targeting teacher-student relationships	(Bosman et al., 2021; Koenen et al., 2021; McGrath & Van Bergen, 2019; Spilt et al., 2012; Yoon, 2002)
39* Self-awareness: Emotions experienced by beginning teachers in relation to themselves (e.g. insecurity)	Social-emotional competence Core elements of interventions targeting teacher-student relationships	(Bosman et al., 2021; Koenen et al., 2021; McGrath & Van Bergen, 2019; Spilt et al., 2012; Yoon, 2002)
40 Self-awareness: Emotions experienced by teachers in relationships with students (e.g. proud, commitment, worry, anger)	Social-emotional competence Core elements of interventions targeting teacher-student relationships	(Bosman et al., 2021; Koenen et al., 2021; McGrath & Van Bergen, 2019; Spilt et al., 2012; Yoon, 2002)
41 Self-awareness: Interpersonal stress (e.g. sources of stress, consequences of stress)	Social-emotional competence	(Carstensen & Klusmann, 2020; Clayback & Williford, 2021; Friedman-Krauss et al., 2014; Whitaker et al., 2015; Yoon, 2002)
43 Self-management: Adaptive coping strategies (e.g. cognitive reappraisal)	Social-emotional competence	(Beltman et al., 2011; Chang, 2009; de Ruiter et al., 2021; Hastings & Brown, 2002; Pilen et al., 2013)
44 Self-management: Maladaptive coping strategies (e.g. neutralizing, avoiding or repressing negative emotions)	Social-emotional competence	(Beltman et al., 2011; Chang, 2009; de Ruiter et al., 2021; Hastings & Brown, 2002; Pilen et al., 2013)
45 Self-management: Mindfulness techniques	Social-emotional competence	(Iancu et al., 2018; Jennings, 2015; Kincade et al., 2020; Roeser et al., 2012)

Appendix B (continued)

Short description	Source	Example references
46 Self-management: Self-compassion	Social-emotional competence	(Iancu et al., 2018; Jennings, 2015; Roeser et al., 2012)
47 Self-management: Emotional labor	Social-emotional competence	(Beltman et al., 2011; Brown et al., 2017; de Ruiter et al., 2021; Wang et al., 2019; Zaretsky & Katz, 2019)
48* Self-management: Reflection techniques and strategies	Social-emotional competence	(Bosman et al., 2021; Kincade et al., 2020; Koenen et al., 2021; Pianta, 1999; Spilt et al., 2012)
49 Professional identity regarding building relationships: moral/philosophical dimension (e.g. ethics of care)	Professional identity	(Aspelin & Jonsson, 2019; Kelchtermans, 1996; Kelchtermans, 2005; O'Connor, 2008; Pillen et al., 2013)
50 Professional identity regarding building relationships: emotional dimension (e.g. teachers' need to belong)	Relational competence	(García-Moya et al., 2019; Kelchtermans & Deketelaere, 2016; O'Connor, 2008; Pillen et al., 2013)
51 Professional identity regarding building relationships: professional dimension (e.g. what is an 'appropriate' relationship?)	Professional identity	(García-Moya et al., 2019; Kelchtermans & Deketelaere, 2016; O'Connor, 2008; Pillen et al., 2013)
52 Professional identity regarding building relationships: performative dimension (teacher-student relationship in function of other pedagogical goals for the student)	Professional identity	(García-Moya et al., 2019; Kelchtermans & Deketelaere, 2016; O'Connor, 2008; Pillen et al., 2013)
53 Ideas or beliefs about the teachers' responsibility for the teacher-student relationship	Extended attachment framework Core elements of interventions targeting teacher-student relationships	(Bosman et al., 2021; Kincade et al., 2020; Koenen et al., 2021; Pianta, 1999; Spilt et al., 2012)
54 Perspective taking; empathize with the inner experiences of the student (with an emphasis on the emotional and social functioning)	Core elements of interventions targeting teacher-student relationships	(Bosman et al., 2021; Kincade et al., 2020; Koenen et al., 2021; Pianta, 1999; Spilt et al., 2012)
55 The student's emotion and stress regulation	Core elements of interventions targeting teacher-student relationships	(Driscoll et al., 2011; Hoogendijk et al., 2020; Kincade et al., 2020; Vancraeyveldt et al., 2015)
56 Praise	Core elements of interventions targeting teacher-student relationships	(Cook et al., 2019; Kincade et al., 2020; Vancraeyveldt et al., 2015)
57 Skills aimed at restoring the relationship	Core elements of interventions targeting teacher-student relationships	(Bosman et al., 2021; Cook et al., 2019; Kincade et al., 2020; Koenen et al., 2021; Spilt et al., 2012)
58 Positive to negative ratio of interactions	Core elements of interventions targeting teacher-student relationships	(Cook et al., 2019; Driscoll et al., 2011; Kincade et al., 2020)
59 One-on-one time	Core elements of interventions targeting teacher-student relationships	(Cook et al., 2019; Driscoll et al., 2011; Kincade et al., 2020; Vancraeyveldt et al., 2015)
60 Student-led activities	Core elements of interventions targeting teacher-student relationships	(Driscoll et al., 2011; Kincade et al., 2020; Vancraeyveldt et al., 2015)
61 Positive greetings and farewells	Core elements of interventions targeting teacher-student relationships	(Cook et al., 2019; Driscoll et al., 2011; Kincade et al., 2020; Vancraeyveldt et al., 2015)
62 Validating and labeling the student's emotions and views	Core elements of interventions targeting teacher-student relationships	(Cook et al., 2019; Driscoll et al., 2011; Kincade et al., 2020; Vancraeyveldt et al., 2015)
63 Physical contact	Relational competence Core elements of interventions targeting teacher-student relationships	(Aspelin & Jonsson, 2019; Kincade et al., 2020)
64 Non-verbal behavior (e.g. gestures)	Relational competence Core elements of interventions targeting teacher-student relationships	(Aspelin & Jonsson, 2019; Driscoll et al., 2011; Kincade et al., 2020; Vancraeyveldt et al., 2015)
65 Posture (e.g. get down to your knees)	Relational competence Core elements of interventions targeting teacher-student relationships	(Aspelin & Jonsson, 2019; Driscoll et al., 2011; Kincade et al., 2020; Vancraeyveldt et al., 2015)

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Appendix B (continued)

Short description	Source	Example references
66 Showing empathy	Relational competence Core elements of interventions targeting teacher-student relationships	(Aspelin & Jonsson, 2019; Cook et al., 2019; Driscoll et al., 2011; Kincade et al., 2020; McGrath & Van Bergen, 2019; Vancraeyveldt et al., 2015)
67 Showing respect	Relational competence Core elements of interventions targeting teacher-student relationships	(Aspelin & Jonsson, 2019; Driscoll et al., 2011; Kincade et al., 2020; Vancraeyveldt et al., 2015)
68 Self-disclosure (sharing personal information)	Core elements of interventions targeting teacher-student relationships Derived from empirical research	(Kincade et al., 2020; Quan-McGimpsey et al., 2011; Taxer et al., 2018)
69 Observation skills	Core elements of interventions targeting teacher-student relationships Derived from empirical research	(Driscoll et al., 2011; Kincade et al., 2020; Vancraeyveldt et al., 2015)
70 Evidence-based interventions targeting the teacher-student relationship	Core elements of interventions targeting teacher-student relationships	(Driscoll et al., 2011; Kincade et al., 2020; Vancraeyveldt et al., 2015)
71 Peer-conflict resolution	Core elements of interventions targeting teacher-student relationships	(Chen, Lindo et al., 2021; Denisenkova & Nisskaya, 2016; Kincade et al., 2020)
72* Sensitive behavior management	Core elements of interventions targeting teacher-student relationships Derived from empirical research	(Bosman et al., 2021; Hoogendijk et al., 2020; Koenen et al., 2021; Spilt et al., 2012; Vancraeyveldt et al., 2015)
73* Offering help (proactive and reactive)	Core elements of interventions targeting teacher-student relationships Derived from empirical research	(Bosman et al., 2021; Hoogendijk et al., 2020; Koenen et al., 2021; Spilt et al., 2012; Vancraeyveldt et al., 2015)
74* Showing interest in the student (their interests, daily life)	Core elements of interventions targeting teacher-student relationships Derived from empirical research	(Cook et al., 2019; Driscoll et al., 2011; Vancraeyveldt et al., 2015)
75* Express positive expectations	Core elements of interventions targeting teacher-student relationships Derived from empirical research	(Cook et al., 2019; Driscoll et al., 2011; Vancraeyveldt et al., 2015)
76* Non-violent communication	Social-emotional competence Relational competence	(Heinrichs, Ziegler, & Warwas, 2021; Rosenberg & Chopra, 2015)
77* Non-prejudice	Social-emotional competence Relational competence	(Aspelin & Jonsson, 2019; Cook et al., 2019; Driscoll et al., 2011; Kincade et al., 2020; McGrath & Van Bergen, 2019; Shifrer, 2013; Vancraeyveldt et al., 2015)
78* Non-discrimination	Social-emotional competence Relational competence	(Aspelin & Jonsson, 2019; Cook et al., 2019; Driscoll et al., 2011; Kincade et al., 2020; McGrath & Van Bergen, 2019; Vancraeyveldt et al., 2015)
79* Competences related to parents	Social-emotional competence Relational competence	(Aspelin & Jonsson, 2019; Chung, et al., 2008; Hughes & Kwok, 2007; Jeon et al., 2021; Lang et al., 2020)

Appendix C

Competencies and ratings of importance

Short description	Additional information	Survey consensus reached	Ratings	Conclusion
1* The importance of teacher-student relationships for the personal well-being of the student	Mainly knowledge: the teacher knows why a positive teacher-student relationship is important for the student (specified: 1 through 6)	Survey 2	All 'need to know'	Need
2 The importance of teacher-student relationships for the social-emotional-motivational development of the student		Survey 2	All 'need to know'	Need
3 The importance of teacher-student relationships for the cognitive development of the student		Survey 2	All 'need to know'	Need
4 The importance of teacher-student relationships for the development of the student's relationships with classmates		Survey 2	All 'need to know', except 1 'nice to know'	Need
5* The importance of teacher-student relationships for the development of the student's relationships with peers	Mainly knowledge: the teacher knows why a positive teacher-student relationship is important for the teacher (specified: 7 through 9)	Survey 3	6 out of 9 'nice to know' 3 'need to know'	Nice
6* The importance of teacher-student relationships for the development of the student's relationships with other adults		Survey 3	All 'nice to know', except 1 'need to know'	Nice
7 The importance of teacher-student relationships for the professional well-being of the teacher		No consensus reached	5 out of 9 'need to know' 3 'nice to know' 1 'not to know'	Nice
8 The importance of teacher-student relationships for the personal well-being of the teacher		Survey 3	6 out of 9 'nice to know' 3 'need to know'	Nice
9 The importance of teacher-student relationships for the teacher's self-efficacy	Mainly knowledge For example: dependent student behavior is valued differently in individualistic as compared to collectivistic cultures	Survey 3	6 out of 9 'nice to know' 3 'need to know'	Nice
10* Cultural differences and similarities in teacher-student relationships		Survey 2	All 'nice to know', except 1 'not to know'	Nice
11 Student characteristics related to the (quality of the) teacher-student relationship		Survey 3	6 out of 9 'nice to know' 2 'need to know' 1 'not to know'	Nice
12 Characteristics of the student's environment (e.g. home environment) related to the (quality of the student-teacher relationship)		Survey 3	7 out of 9 'nice to know' 1 'not to know' 1 'need to know'	Nice
13 Teacher characteristics related to the (quality of the) teacher-student relationship	Mainly knowledge: the teacher knows what characteristics of themselves (or other teachers) are related to (the quality of) their relationships with students	Survey 3	7 out of 9 'nice to know' 2 'need to know'	Nice
14 Characteristics of the school or class related to the (quality of the) teacher-student relationship	Mainly knowledge: the teacher knows what characteristics of the school or class are related to (the quality of) teacher-student relationships	Survey 3	All 'nice to know', except 1 'need to know'	Nice
15* Building relationships with students with special educational needs	Knowledge, yet also skill: remarks or specific skills related to building relationships with students with special educational needs For example: establishing contact with a blind student, establishing contact with a student who does not or barely uses language	Survey 2	All 'need to know', except 1 'nice to know'	Need
16 Building relationships with students with a physical disability	Further specification of competence 15	Survey 2	All 'nice to know', except 1 'need to know'	Nice
17* Building relationships with students with a (mild) mental disability	Further specification of competence 15	Survey 2	All 'nice to know', except 1 'need to know'	Nice
18 Building relationships with students with learning problems or disorders	Further specification of competence 15	Survey 3	7 out of 9 'nice to know' 2 'need to know'	Nice
19 Building relationships with students with behavioral or emotional problems or disorders	Further specification of competence 15	Survey 3	6 out of 9 'nice to know' 2 'need to know' 1 'not to know'	Nice

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Appendix C (continued)

	Short description	Additional information	Survey consensus reached	Ratings	Conclusion
20	Building relationships with students with attachment problems or disorders	Further specification of competence 15	Survey 3	6 out of 9 'nice to know' 3 'not to know'	Nice
21*	Building relationships with students from different (social-economic, cultural, religious ...) backgrounds	Knowledge, yet also skill; remarks or specific skills related to building relationships with students with a different background (than the teacher) For example: establishing contact with a student who does not speak the instructional language	Survey 2	All 'need to know', except 1 'nice to know'	Need
22	How the teacher-student relationship relates to other concepts (e.g. class climate, behavior management)	Mainly knowledge: How the teacher-student relationship relates to other concepts on the teacher's pedagogical practices and class climate For example: the relationship between one-on-one teacher-student relationships and classroom-level relationships For example: how a positive teacher-student relationship can facilitate other tasks such as behavior management	Survey 3	6 out of 9 'nice to know' 2 'need to know' 1 'not to know'	Nice
23	Attachment theory: Background of attachment theory (e.g. Bowlby, Harlow, Ainsworth)	Knowledge	Survey 2	All 'nice to know', except 1 'not to know'	Nice
24	Attachment theory: Central concepts such as secure base and safe haven and autonomous exploration	Mainly knowledge	Survey 2	All 'need to know', except 1 'nice to know'	Need
25	Attachment theory: Similarities and differences between parent-child and teacher-student relationships	Mainly knowledge	Survey 3	All 'nice to know', except 1 'need to know'	Nice
26	Attachment theory: Role of the teacher as an ad hoc attachment figure	Mainly knowledge	Survey 3	6 out of 9 'nice to know' 3 'need to know'	Nice
27	Attachment theory: Describing relationship quality in terms of closeness, conflict and dependency	Mainly knowledge	Survey 4	6 out of 9 'need to know' 3 'nice to know'	Need
28	Attachment theory: Teachers' mental representations or internal working models of teacher-student relationships *	Mainly knowledge (*internalized beliefs and feelings about the other, the self in relation to the other, and the self-other relationship)	Survey 2	All 'need to know', except 1 'nice to know'	Need
29	Attachment theory: The student's mental representations or internal working models *	Mainly knowledge (* internalized beliefs about the educator/teacher as (un)trustworthy and (un)available, and the self as (un)worthy of receiving attention and love)	Survey 3	7 out of 9 'nice to know' 1 'need to know' 1 'not to know'	Nice
30	Attachment theory: Teacher sensitivity	Knowledge, yet also skill	Survey 4	7 out of 9 'need to know' 2 'nice to know'	Need
31	Self-determination theory: Importance of teacher-student relationship from a motivational perspective	Mainly knowledge	Survey 2	All 'need to know', except 1 'nice to know'	Need
32	Self-determination theory: Need for Autonomy, Belonging and Competence (ABC)	Mainly knowledge	Survey 2	All 'need to know', except 1 'nice to know'	Need
33	Self-determination theory: Supporting students' ABC needs via autonomy support, involvement, structure	Knowledge, yet also skill	Survey 2	All 'need to know', except 1 'nice to know'	Need
34	Theory of Interpersonal Teacher Behavior: Theoretical background (systems communication theory, Leary Model, Wubbels & Brekelmans)	Mainly knowledge	Survey 2	All 'nice to know', except 1 'not to know'	Nice
35	Theory of Interpersonal Teacher Behavior: Describing interactions between partners based on two dimensions Dominance (dependent versus dominant) and Affiliation (hostile versus friendly)	Mainly knowledge	Survey 3	7 out of 9 'nice to know' 1 'need to know' 1 'not to know'	Nice
36*	Theory of Interpersonal Teacher Behavior: Interpersonal circle and 8 possible positions	Mainly knowledge	Survey 3	7 out of 9 'nice to know' 2 'need to know'	Nice
37	Theory of Interpersonal Teacher Behavior: Teachers' behavior elicits complementary behavior from the student and vice versa ((a)symmetry and (non)complementarity in student-teacher interactions)	Mainly knowledge	Survey 3	6 out of 9 'nice to know' 3 'need to know'	Nice
38	Self-awareness: Emotions experienced by beginning teachers in relation to their tasks as a teacher (e.g. helplessness)	Skill For example: Recognizing these emotions within yourself, for example during the internship	Survey 3	6 out of 9 'nice to know' 3 'need to know'	Nice
39*	Self-awareness: Emotions experienced by beginning teachers in relation to themselves (e.g. insecurity)		Survey 3	6 out of 9 'nice to know'	Nice

Appendix C (continued)

Short description	Additional information	Survey consensus reached	Ratings	Conclusion
	Skill For example: Recognizing these emotions within yourself, for example during the internship		2 'need to know' 1 'not to know'	
40 Self-awareness: Emotions experienced by teachers in relationships with students (e.g. proud, commitment, worry, anger)	Skill For example: Recognizing these emotions within yourself, for example during the internship	No consensus reached	5 out of 9 'need to know'	Nice
41 Self-awareness: Interpersonal stress (e.g. sources of stress, consequences of stress)	Knowledge and skill For example: Knowledge about frequent sources and consequences of stress For example: Recognizing stress signals within yourself, for example during the internship	Survey 3	4 'nice to know' 6 out of 9 'nice to know' 3 'need to know'	Nice
42 Self-awareness: Ideas and thoughts about individual students (e.g. attributions of problem behavior)	Skill For example: Awareness of and reflection on your own ideas and opinions of students, for example about the cause of (problem)behavior during the internship	Survey 2	All 'need to know', except 1 'nice to know'	Need
43 Self-management: Adaptive coping strategies (e.g. cognitive reappraisal)	Knowledge and skill Knowledge about adaptive coping strategies For example: Being able to apply adaptive coping strategies during the internship	Survey 3	6 out of 9 'nice to know' 3 'need to know'	Nice
44 Self-management: Maladaptive coping strategies (e.g. neutralizing, avoiding or repressing negative emotions)	Knowledge and skill Knowledge about maladaptive coping strategies For example: Recognizing the use of maladaptive strategies, for example during the internship	Survey 3	6 out of 9 'nice to know' 2 'need to know' 1 'not to know'	Nice
45 Self-management: Mindfulness techniques	Knowledge and skill Knowledge about mindfulness For example: Being able to apply mindfulness techniques, for example during the internship	Survey 2	All 'nice to know', except 1 'not to know'	Nice
46 Self-management: Self-compassion	Attitude/skill For example: Coping with challenges and fallbacks in a positive way, for example during the internship	Survey 3	6 out of 9 'nice to know' 3 'need to know'	Nice
47 Self-management: Emotional labor	Knowledge and skill For example: Knowledge of emotional labor (what is this? What strategies exist?) For example: Recognizing in your own practice what strategies you use, for example during the internship	Survey 3	6 out of 9 'nice to know' 3 'not to know'	Nice
48* Self-management: Reflection techniques and strategies	Knowledge and skill Knowledge about reflection techniques (e.g. Korthagen) Being able to apply reflection techniques, for instance to experiences during the internship	Survey 4	6 out of 9 'nice to know' 3 'need to know'	Nice
49 Professional identity regarding building relationships: moral/philosophical dimension (e.g. ethics of care)	Attitude For example: what is my 'duty' as a teacher?	Survey 2	All 'nice to know', except 1 'need to know'	Nice
50 Professional identity regarding building relationships: emotional dimension (e.g. teachers' need to belong)	Attitude For example: are relationships with students important to me as a person?	Survey 3	6 out of 9 'nice to know' 3 'need to know'	Nice
51 Professional identity regarding building relationships: professional dimension (e.g. what is an 'appropriate' relationship?)	Attitude For example: how far can you go in a relationship with you students, e.g. do you offer them help outside the classroom? How can you guard your own needs and interests?	Survey 4	6 out of 9 'need to know' 3 'nice to know'	Need
52 Professional identity regarding building relationships: performative dimension (teacher-student relationship in function of other pedagogical goals for the student)	Attitude For example: why is it important for me to build relationships with students? Is this about the relationship in itself, or because a positive relationships makes it easier to fulfill other tasks such as behavior management?	Survey 3	All 'nice to know', except 1 'need to know'	Nice
53 Ideas or beliefs about the teachers' responsibility for the teacher-student relationship	Attitude For example: can I, as a teacher, do much to build a relationship? Is the relationship with a student strongly influenced by the student's personality or other student characteristics?	Survey 2	All 'need to know'	Need
54 Perspective taking; empathize with the inner experiences of the student (with an emphasis on the emotional and social functioning)	Skill	Survey 3	7 out of 9 'need to know' 2 'nice to know'	Need
55 The student's emotion and stress regulation	Skill	Survey 2	All 'need to know'	Need
56 Praise	Skill	Survey 2	All 'need to know'	Need
57 Skills aimed at restoring the relationship	Skill	Survey 3	6 out of 9 'nice to know'	Nice

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Appendix C (continued)

Short description	Additional information	Survey consensus reached	Ratings	Conclusion
58 Positive to negative ratio of interactions	Skill	Survey 2	3 'need to know' All 'need to know', except 1 'nice to know'	Need
59 One-on-one time	Skill	Survey 2	All 'need to know'	Need
60 Student-led activities	Skill	Survey 3	6 out of 9 'nice to know' 2 others 'need to know' 1 'not to know'	Nice
61 Positive greetings and farewells	Skill	Survey 2	All 'need to know'	Need
62 Validating and labeling the student's emotions and views	Skill	Survey 2	All 'need to know'	Need
63 Physical contact	Skill	Survey 3	6 out of 9 'nice to know' 2 'need to know'	Nice
64 Non-verbal behavior (e.g. gestures)	Skill	Survey 2	All 'need to know', except 1 'nice to know'	Need
65 Posture (e.g. get down to your knees)	Skill	Survey 2	All 'need to know', except 1 'nice to know'	Need
66 Showing empathy	Skill	Survey 2	All 'need to know'	Need
67 Showing respect	Attitude/Skill	Survey 2	All 'need to know'	Need
68 Self-disclosure (sharing personal information)	Skill	Survey 3	7 out of 9 'nice to know' 2 'need to know'	Nice
69 Observation skills	Skill	Survey 3	7 out of 9 'need to know' 2 'nice to know'	Need
70 Evidence-based interventions targeting the teacher-student relationship	Skill	Survey 3	All 'nice to know'	Nice
71 Peer-conflict resolution	Skill	Survey 2	All 'need to know', except 1 'nice to know'	Need
72* Sensitive behavior management	Skill	Survey 4	8 out of 9 'need to know' 1 'nice to know'	Need
73* Offering help (proactive and reactive)	Skill	Survey 2	All 'need to know'	Need
74* Showing interest in the student (their interests, daily life)	Skill	Survey 2	All 'need to know'	Need
75* Express positive expectations	Skill	Survey 2	All 'need to know', except 1 'nice to know'	Need
76* Non-violent communication	Skill	Survey 2	All 'need to know'	Need
77* Non-prejudice	Attitude	Survey 2	All 'need to know', except 1 'nice to know'	Need
78* Non-discrimination	Attitude	No consensus reached	5 out of 9 'need to know' 4 'nice to know'	Nice
79* Competences related to parents	These competences are only indirectly related to the teacher-student relationship. For example: parent involvement, cooperation with parents, shared responsibilities and care, get more information on the student's needs through the parents ...	Survey 2	All 'need to know'	Need

Note: Items added by panel members are indicated with an asterisk (*).

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